Distributions and Maximum Likelihood

Section 1: Discrete Random Variables and Bernoulli Likelihood

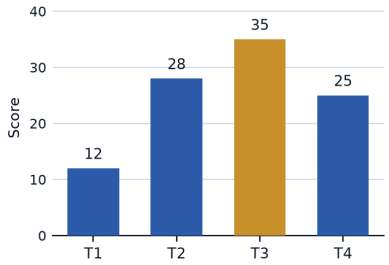
1 In a binary click-through model, an ad is clicked $k = 40$ times out of $n = 200$ impressions. What is the Maximum Likelihood Estimate $\hat{p}_{MLE}$ of the click probability?

a. 0.20

b. 0.25

c. 0.16

d. 0.40



2 What is the variance of a Bernoulli random variable with success probability $p = 0.3$?

a. 0.21

b. 0.30

c. 0.70

d. 0.49

3 Show how maximizing the Bernoulli log-likelihood relates to the binary cross-entropy loss function used in neural networks.

a. Maximizing Bernoulli log-likelihood is mathematically identical to minimizing binary cross-entropy

b. Cross-entropy is the reciprocal of the log-likelihood

c. They only match when $p = 0.5$

d. Cross-entropy ignores negative classes

Section 2: Continuous Gaussian Normal Distributions

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4 A feature vector component follows $X \sim \mathcal{N}(50, 16)$. What are its mean μ and standard deviation σ?

a. $\mu = 50, \quad \sigma = 4$

b. $\mu = 50, \quad \sigma = 16$

c. $\mu = 16, \quad \sigma = 50$

d. $\mu = 25, \quad \sigma = 4$

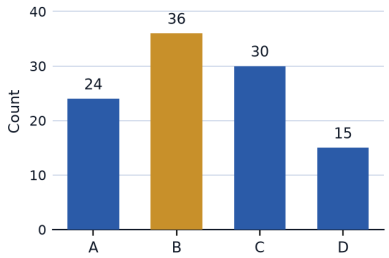
5 By the empirical $(68 - 95 - 99.7)$ rule, approximately what percentage of samples from $\mathcal{N}(\mu, \sigma^2)$ fall within the interval $(\mu - 2\sigma, \mu + 2\sigma)$?

a. 95.4%

b. 68.2%

c. 99.7%

d. 50.0%



6 Standardize the value $x = 62$ from the distribution $\mathcal{N}(50, 16)$ into a standard normal z-score.

a. $z = 3.0$

b. $z = 0.75$

c. $z = 12$

d. $z = 1.5$

Section 3: The Likelihood Function and Log-Likelihood Formulation

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7 Why is the log-likelihood $\ln L(\theta) = \sum_{i=1}^{n} \ln p(x_i|\theta)$ preferred over raw likelihood in machine learning optimization?

a. It transforms products of probabilities into sums, preventing numerical underflow and simplifying derivatives

b. It converts non-convex problems into concave problems unconditionally

c. It eliminates the need for normalization

d. It ensures the sample mean is zero

8 For n independent samples from $\mathcal{N}(\mu, \sigma^2)$, what is the analytical expression for the log-likelihood?

a. $-\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$

b. $-\frac{n}{2}\ln(2\pi) - \sum_{i=1}^{n}(x_i - \mu)$

c. $\frac{n}{2\sigma^2}\sum(x_i - \mu)^2$

d. $-\frac{1}{\sigma^2}\sum(x_i - \mu)$

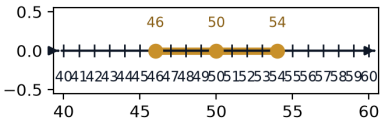
9 Given 5 observations $\{2, 4, 4, 6, 9\}$, compute the sample mean $\bar{x}$, which is the Gaussian MLE $\hat{\mu}_{MLE}$.

a. 5.0

b. 4.0

c. 4.6

d. 5.5

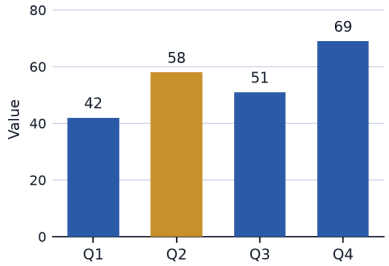


Section 4: Properties of Estimators: Bias and Consistency

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10 Why is the Maximum Likelihood Estimator for variance $\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum(x_i - \bar{x})^2$ biased for finite samples?

a. Its expected value is $\frac{n-1}{n}\sigma^2 < \sigma^2$, underestimating true variance due to using sample mean $\bar{x}$



b. It only works for standard normal data

c. Its expected value approaches infinity

d. It ignores outliers

11 What is Bessel's correction to make the sample variance estimator unbiased?

a. Multiply by $\frac{n}{n-1}$

b. Multiply by $\frac{n-1}{n}$

c. Add $\frac{1}{n}$

d. Subtract the mean

12 What does it mean for an estimator $\hat{\theta}_n$ to be asymptotically consistent?

a. As sample size $n \to \infty$, $\hat{\theta}_n$ converges in probability to the true parameter θ

b. The estimator has zero variance for all sample sizes

c. The estimator is unbiased for $n = 1$

d. The estimator follows a student-t distribution

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